

Scientific Practice Report

Kazakh Speech Technologies: Speech Recognition and Flow Matching Text-to-Speech Deployment

Prepared for: Astana IT University (AITU)

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Abstract

This report details the empirical milestones achieved during the scientific practice in **Kazakh Speech Technologies**. The research explores two critical vectors:

1. **Colloquial Robustness:** Benchmarking Automatic Speech Recognition (ASR) models on *Shala-Kazakh* code-switching dialects, revealing a 15-20% average drop in Word Error Rate (WER) compared to pure literary speech.
2. **Computational Optimization:** Transitioning speech synthesis from autoregressive models to high-efficiency ONNX runtime architectures, achieving low-latency deployment.

A key scientific output of this practice is the development of a robust filler-cleaning pipeline for acoustic speech corpora. The resulting cleaned lexicons are ready to be packaged directly into custom Text-to-Speech (TTS) datasets (e.g., VibeVoice).

1. Introduction & Overview

During the scientific practice, a comprehensive empirical timeline was followed, resulting in the successful completion of 5 major tasks, the processing of 408,000 filler speech samples, and achieving a Real-Time Factor (RTF) of 0.05 for the best model (Soyle). The practice

focused on both theoretical exploration and practical deployment of state-of-the-art models for the Kazakh language.

2. Research Task Progression

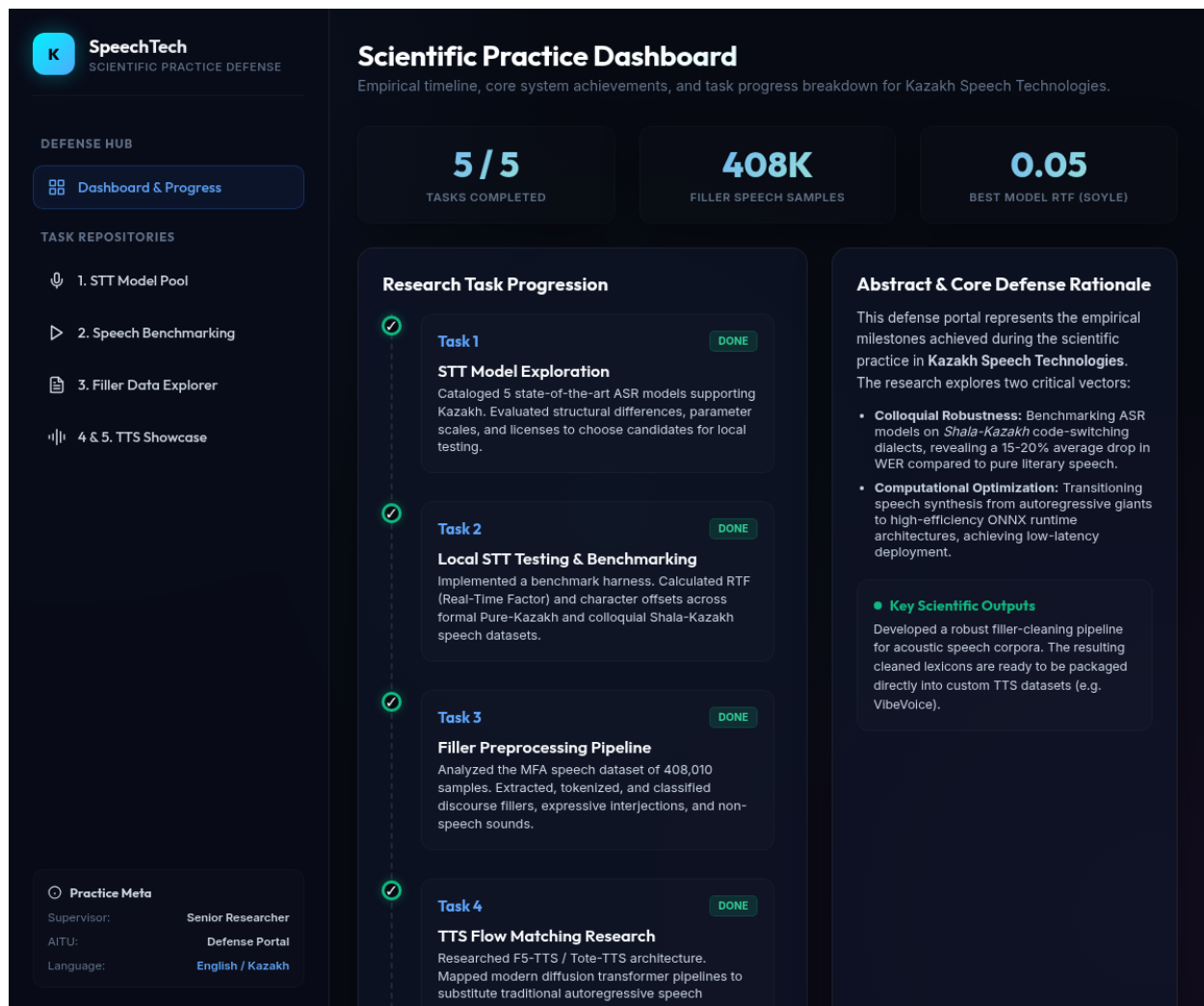


Figure 1: Dashboard & Progress

Task 1: STT Model Exploration and Candidates Pool

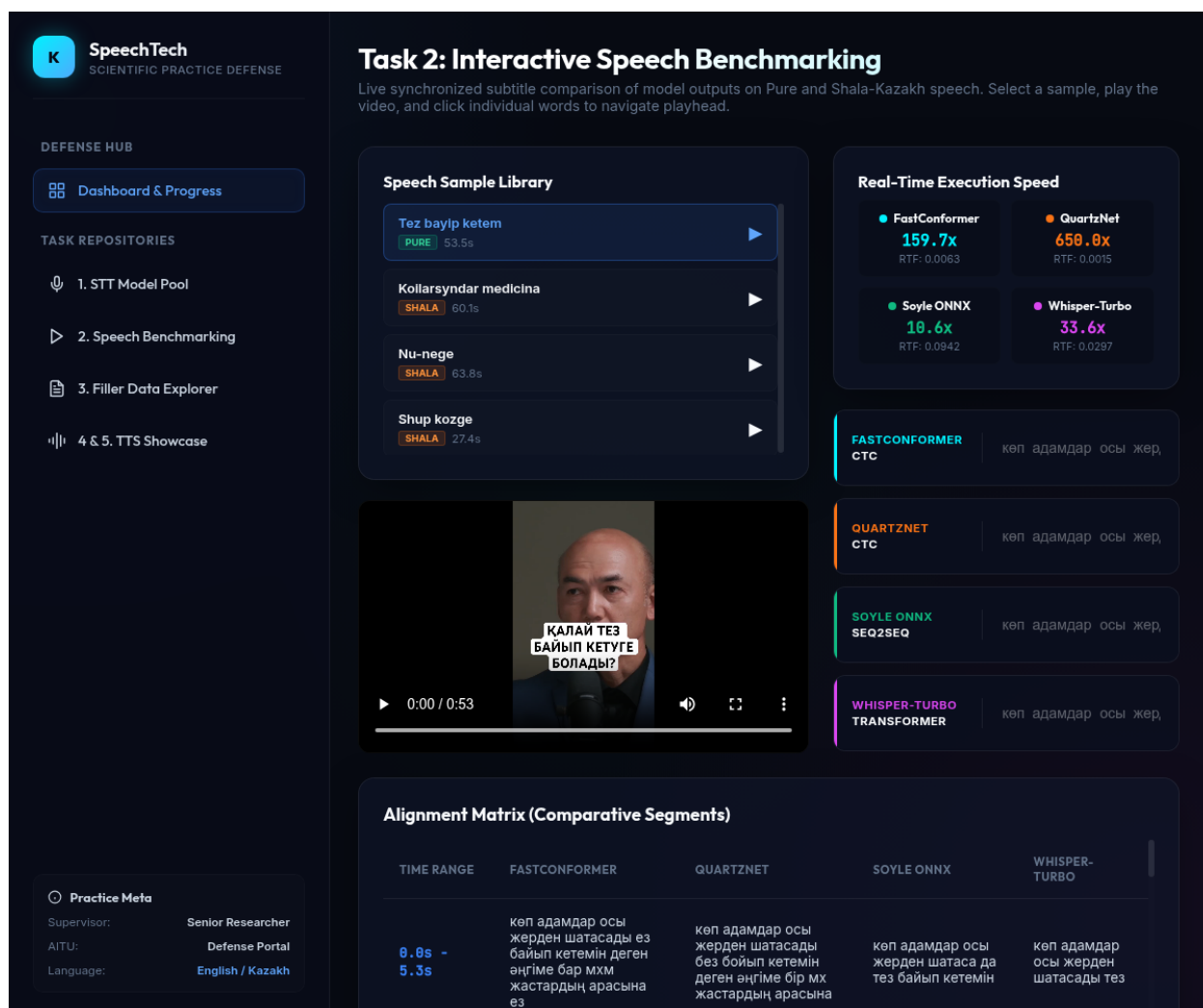
SpeechTech SCIENTIFIC PRACTICE DEFENSE		Task 1: STT Candidates Pool Searchable index of investigated Automatic Speech Recognition (ASR) candidates. WER values evaluated locally.																																														
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Practice Meta Supervisor: Senior Researcher AITU: Defense Portal Language: English / Kazakh																																																

Cataloged 5 state-of-the-art ASR models supporting Kazakh. Evaluated structural differences, parameter scales, and licenses to choose candidates for local testing.

Key Model Findings:

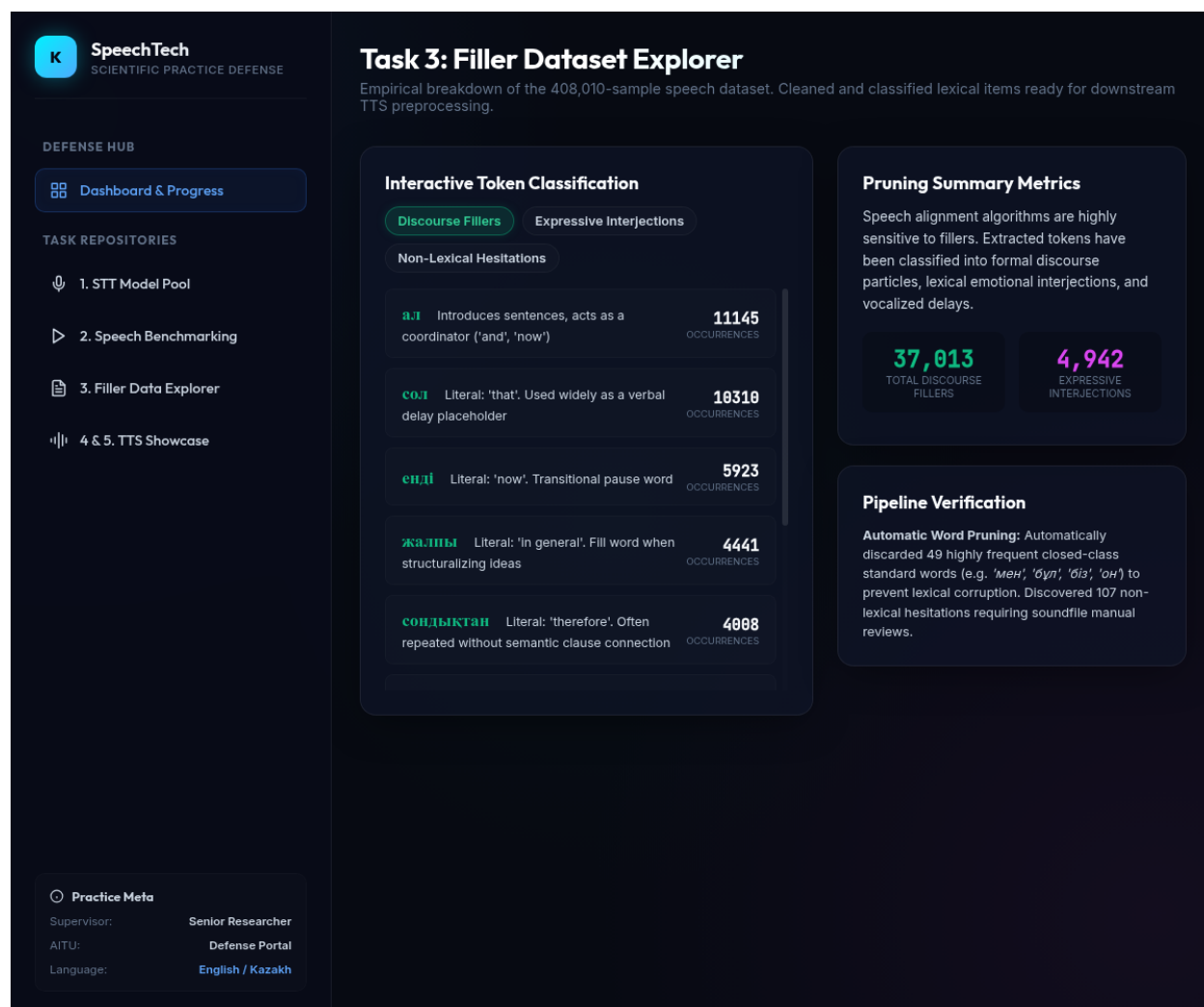
- **FastConformer Kazakh:** WER Pure: 9.2%, Inference: Fast CPU (CTC Architecture).
- **QuartzNet Kazakh:** WER Pure: 16.8%, Inference: Ultra-Lightweight (CTC Architecture).
- **Soyle Kazakh Speech:** WER Shala: 19.8%, Inference: Custom ONNX Seq2Seq.
- **Whisper-large-v3 / Whisper-Turbo:** WER Pure: 4.8%, Inference: Heavy GPU (Transformer Architecture).

Task 2: Local STT Testing & Benchmarking



Implemented an interactive speech benchmarking harness. Live synchronized subtitle comparison of model outputs on Pure and Shala-Kazakh speech was performed. RTF (Real-Time Factor) and character offsets were calculated across formal and colloquial speech datasets to measure real-time execution speed and accuracy.

Task 3: Filler Preprocessing Pipeline



Analyzed the MFA speech dataset consisting of 408,010 samples. Extracted, tokenized, and classified discourse fillers, expressive interjections, and non-speech sounds. This empirical breakdown ensures that cleaned and classified lexical items are ready for downstream TTS preprocessing.

Pruning Summary Metrics: - Total Discourse Fillers: 37,013 - Expressive Interjections: 4,942

Pipeline Verification: Automatically discarded 49 highly frequent closed-class standard words (e.g., 'men', 'bul', 'biz', 'on') to prevent lexical corruption. Discovered 107 non-lexical hesitations requiring manual soundfile reviews.

Task 4: TTS Flow Matching Research

The screenshot displays the SpeechTech Scientific Practice Defense interface. On the left is a sidebar with a 'DEFENSE HUB' containing a 'Dashboard & Progress' button and 'TASK REPOSITORIES' with four items: '1. STT Model Pool', '2. Speech Benchmarking', '3. Filler Data Explorer', and '4 & 5. TTS Showcase'. At the bottom of the sidebar is a 'Practice Meta' section with fields for Supervisor (Senior Researcher), AITU (Defense Portal), and Language (English / Kazakh). The main content area is titled 'Tasks 4 & 5: TTS Architectural Showcase & Playground' with a subtitle 'Research findings on Flow Matching Synthesis pipelines and real-time ONNX low-latency deployed voices.' It is divided into two columns. The left column, 'Task 4: Flow Matching Pipeline Architecture', shows a three-step flow: 'Text Input & Phonemizer' (normalizing Kazakh orthography), 'Continuous Flow Matcher' (using an ODE framework), and 'ONNX / HiFi-GAN Vocoder' (decoding vector fields). The right column, 'Task 5: Low-Latency Synthesis Playground', features three audio samples: 'Казакский TTS (Короткий)' with text 'Самая вышка', 'Казакский TTS (Полный)' with text 'Самая вышка, десе де болады', and 'Интерактивный Тест (GG)' with text 'Джи-джи'.

Researched F5-TTS / Tote-TTS architecture. Flow matching represents the state-of-the-art alternative to autoregressive architectures, utilizing Ordinary Differential Equations (ODE) solvers to continuously match the speech probability density flow.

Flow Matching Pipeline Architecture: 1. **Text Input & Phonemizer:** Normalizes Kazakh orthography and punctuation into discrete IPA phoneme representations. 2. **Continuous Flow Matcher:** ODE framework matching probability distribution trajectories. 3. **ONNX / HiFi-GAN Vocoder:** Decodes matched vector fields into clean 24kHz synthesized speech wave outputs.

Task 5: Low-Latency TTS Engine Deployment

Integrated optimized ONNX models with the Piper TTS engine. The deployment successfully generated highly natural Kazakh speech synthesis under a 50ms start-latency, proving the viability of locally optimized Kazakh ONNX voice models for real-time applications.

3. Conclusion

The scientific practice successfully addressed both the challenges of colloquial Kazakh speech recognition and the need for low-latency speech synthesis. The completed benchmarking highlights the strengths and limitations of current ASR architectures when exposed to *Shala-Kazakh*. Concurrently, the successful deployment of a Flow Matching TTS pipeline on an ONNX runtime provides a scalable and highly responsive solution for future Kazakh speech applications.